Note 17.

Linear differential equations of second order with constant coefficients

The equation

$$y'' = -k^2 y \tag{17.1}$$

describes the *oscillations* of a particle on a line around the origin $y = 0$, when a force proportional to its displacement pulls it back to its equilibrium position at $y = 0$.[1]

This is an example of a *second order* equation, and could we viewed as a *system of first order equations*: Introducing the unknowns $y_1 = y$ and $y_2 = y'$ we could rewrite this equation as

$$y_1' = y_2\,, \qquad y_2' = -k^2 y_1\,. \tag{17.2}$$

We can think of $\vec{y} = (y_1, y_2)$ as a point in the plane, and view the solution $\vec{\varphi}(t)$ as a point moving through 2-dimensional space $\mathbb{R}^2$ with velocity $\vec{v}$:

$$\vec{\varphi}'(t) = \vec{v}(\vec{\varphi}(t)) \tag{17.3}$$

Here $\vec{v}(y_1, y_2) = (y_2, -k^2 y_1)$ is a **vectorfield**.

Exercise 17.1. A *vectorfield* in $\mathbb{R}^2$, in comparison to a *direction field* which has a line attached to each point, has in addition to the slope also a *magnitude* at each point. In fact, with $\vec{v}(y_1, y_2) = (v_1(y_1, y_2), v_2(y_1, y_2))$ the magnitude at the point (y_1, y_2) is $|v| = \sqrt{v_1^2 + v_2^2}$. Sketch the vectorfield $\vec{v}$ of (17.2).

Exercise 17.2. Show that in the case $k = 1$ concentric circles are solutions to (17.2). The fact that $y_1^2/2 + y_2^2/2$ is constant along the solution curve can be interpreted as the law of *conservation of energy.* What is the situation if $k \neq 1$?

While recasting a second order equation as a first order system is a very fruitful point of view, we will not adopt this approach here, and study directly the *linear differential equation of second order with constant coefficients*:

$$y'' + ay' + by = 0 \tag{17.4}$$

More generally, a **linear differential equations of second order** is an expressions of the form

$$y'' + g(x)y' + h(x)y = r(x) \tag{17.5}$$

In the case that $r(x)$ vanishes identically we say the equation is *homogeneous*, otherwise *inhomogeneous.*

[1] For the given equation, this relationship is *linear*, and is then also referred to as *Hooke's law* in physics.

17.1. Existence of solutions by inspection

We consider first the equation (17.4) in the case that $a = 0$.

The equation $y'' = 0$. In this special case, when both coefficients $a = b = 0$, we see immediately by integration that any solution that satisfies this equation has constant derivative $y' = c_1$, and so integrating again

$$y(x) = c_1 x + c_2 \tag{17.6}$$

with constant c_2. Conversely, for any numbers $c_1, c_2 \in \mathbb{R}$, the function given by (17.6) is a solution, so we have found all solutions in this case.

The equation $y'' + by = 0$, when $b < 0$. Since $b < 0$, we can write $b = -k^2$ for some $k > 0$, and the differential equation takes the form

$$y'' = k^2 y\,. \tag{17.7}$$

We immediately verify that $y(x) = e^{kx}$ is a solution, and another is $y(x) = e^{-kx}$. Therefore also linear combinations of these are solutions, and we conclude: For any constants $c_1, c_2 \in \mathbb{R}$

$$y(x) = c_1 e^{kx} + c_2 e^{-kx} \tag{17.8}$$

is a solution. We will prove below that these are in fact *all* solutions in this case.

The equation $y'' + by = 0$, when $b > 0$. In this case let us again write $b = k^2$, then the equation (17.4) takes the form

$$y''(x) = -k^2 y\,. \tag{17.9}$$

One may recognize that this relation is satisfied by the function $y(x) = \cos(kx)$, and also $y(x) = \sin(kx)$. We find again a general solution by forming a linear combination:

$$y(x) = c_1 \cos(kx) + c_2 \sin(kx) \tag{17.10}$$

This does *not* show that any solution of (17.9) is of this form, but we have actually already given a proof of that in Note 12 in the case $k = 1$.

Exercise 17.3. Can you generalise the argument we have given in Lemma 12.2 to show that *any* solution to (17.9) is of the form (17.10)?

The case $a \neq 0$. The cases considered above actually cover the case $a \neq 0$ as well, in the sense that we can *reduce* the problem (17.4) to the problem with $a = 0$.

The idea is to consider solutions of (17.4) of the form

$$y(x) = u(x)v(x)\,. \tag{17.11}$$

A simple calculation gives that

$$y'' + ay' + by = (v'' + av' + bv)u + (2v' + av)u' + vu'' \tag{17.12}$$

and so by choosing v such that $v' = -av/2$ — which can evidently be arranged by choosing $v(x) = e^{-ax/2}$ — the coefficient to the u' term disappears, and with this choice the equation reduces to

$$y'' + ay' + by = \Big(u'' + \frac{4b - a^2}{4}u\Big)v\,. \tag{17.13}$$

Since $v(x) = e^{-ax/2} \neq 0$, we have shown:

Proposition 17.1. *Let $y(x) = u(x)\exp(-ax/2)$. Then the function y satisfies* (17.4) *if and only if the function u satisfies the differential equation*

$$u'' + \frac{4b - a^2}{4}u = 0 \tag{17.14}$$

17.2. General form of the solutions

Theorem 17.2. *Let $b \in \mathbb{R}$. Every solution $\varphi(x)$ of the equation*

$$y'' + by = 0 \tag{17.15}$$

is a linear combination of two functions f_1, f_2, where

- *when $b = 0$: $f_1(x) = 1$, $f_2(x) = x$*
- *when $b < 0$: set $b = -k^2$ and $f_1(x) = e^{kx}$, $f_2(x) = e^{-kx}$*
- *when $b > 0$: set $b = k^2$ and set $f_1(x) = \cos(kx)$, and $f_2(x) = \sin(kx)$.*

In other words, for every solution $\varphi(x)$ of the equation (17.15) *there are constants $c_1, c_2 \in \mathbb{R}$ so that*

$$\varphi(x) = c_1 f_1(x) + c_2 f_2(x)\,. \tag{17.16}$$

Proof. Let $\varphi(x)$ be a solution to (17.15), and let $f_1(x)$, and $f_2(x)$ be chosen as in the statement of the theorem, depending on the value of b. Then also

$$\psi(x) = \varphi(x) - c_1 f_1(x) - c_2 f_2(x) \tag{17.17}$$

is a solution to (17.15).

Let us choose the constants c_1, and c_2 so that $\psi(0) = 0$, and $\psi'(0) = 0$. This amounts to solving the equations:

$$\varphi(0) = c_1 f_1(0) + c_2 f_2(0) \tag{17.18a}$$
$$\varphi'(0) = c_1 f_1'(0) + c_2 f_2'(0) \tag{17.18b}$$

In the case $b = 0$, these equations reduce to

$$\varphi(0) = c_1 \qquad \varphi'(0) = c_2\,. \tag{17.19}$$

Now in the case $b < 0$, these two equations read:

$$\begin{aligned}\varphi(0) =& c_1 + c_2\\ \varphi'(0) =& c_1 k - c_2 k\end{aligned}$$

which we can solve to find

$$\begin{aligned}c_1 =& \frac{\varphi(0) + \varphi'(0)/k}{2}\\ c_2 =& \frac{\varphi(0) - \varphi'(0)/k}{2}\end{aligned}$$

Similary in the case $b > 0$.

Thus it remains to show that the only solution to the initial value problem

$$y'' + by = 0 \qquad y(0) = 0 \qquad y'(0) = 0 \tag{17.20}$$

is the trivial solution $y = 0$.

In the case $b = 1$ we have already demonstrated this in Note 12; see Lemma 12.2.

Exercise 17.4. Generalize this argument to the case $b > 0$.

In the case $b = 0$ this follows from integrating twice.

Exercise 17.5. Show that if f is twice differentiable and satisfies $f''(x) = 0$, and moreover $f(0) = f'(0) = 0$, then $f(x) = 0$.

The case $b < 0$ is covered by the more general considerations leading to Theorem 17.2 below; (see the **additional notes** to Module VI).

□

Exercise 17.6. In view of Proposition 17.1 it is now also possible to state a theorem that gives a characterisation of all solutions to (17.4), for any values of $a, b \in \mathbb{R}$, depending on the sign of the so-called **discriminant** of the differential equation (17.4):

$$\Delta = a^2 - 4b\,. \tag{17.21}$$

State it!

Problems

1. Find explicitly the solution to the following initial value problem:
   a) $y'' + ky = 0$, $y(0) = 0$, $y'(0) = y_1$, where $k > 0$, and $y_1 \in \mathbb{R}$ are fixed constants.
   b) $y'' + ay' = 0$, $y(0) = 1$, $y'(0) = 0$, where $a \in \mathbb{R}$ is fixed.

2. Show that the solution (17.16) in the case $b > 0$ can also be written as

$$\varphi(x) = c\sin(kx + \delta) \tag{17.22}$$

where $c = \sqrt{c_1^2 + c_2^2}$, and δ is to be determined.

3. Find all solutions to the following differential equations:
   a) $y'' + 4y' = 0$
   b) $y'' + 2y' + y = 0$

4. Find all values of the constant k so that $y'' + ky = 0$ has a non-trivial solution $y = \varphi_k(x)$ for which $\varphi_k(0) = \varphi_k(1) = 0$.